When Does Deep Forecasting Stop Paying?
A Benchmark for Demand Forecasting Under Data Scarcity

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September 10, 2026

Abstract

Modern deep forecasting models are evaluated almost exclusively in the enterprise-scale regime: thousands of series, years of clean daily history, and curated preprocessing. The businesses that most need demand forecasting operate nowhere near that regime. A small retailer typically holds six to eighteen months of history across dozens of stock-keeping units, punctuated by stockouts, reporting gaps and long runs of zero demand. Whether the accuracy advantages reported for deep architectures survive this transition is, to our knowledge, untested in a controlled way.

We introduce a scarcity sampling protocol that turns data scarcity into a measurable experimental factor rather than a property of whichever dataset happens to be at hand. The protocol varies history length, series count, missingness rate and missingness shape on an explicit grid, stratifies series by the Syntetos–Boylan demand-pattern quadrants, and emits a manifest that makes every cell exactly reproducible. Two design features are central: unobserved periods are held distinct from observed zeros throughout, and series selection is seeded independently of the scarcity axes so that comparisons across history lengths are paired rather than confounded.

[PENDING EXPERIMENT] The empirical crossover results, the model comparison, and the attribution stability analysis are not yet available. This draft presents the protocol and its validation. No performance numbers appear anywhere in this document, and none should be inferred.

1 Introduction

Forecasting research has converged on a small number of large benchmarks. The M5 competition [Makridakis et al., 2022] provides 30,490 item–store series with 1,941 daily observations each; deep architectures such as N-BEATS [Oreshkin et al., 2020] and the Temporal Fusion Transformer [Lim et al., 2021] report their headline results on data of this scale. The implicit claim carried by these results is that learned representations generalize across a large pool of related series, and that the pooling is what buys the accuracy.

That claim, if true, has an immediate corollary that the literature rarely states: the advantage should decay as the pool shrinks. A firm with forty SKUs and nine months of history is not a scaled-down M5. It is a qualitatively different estimation problem, in which the number of free parameters in a transformer may exceed the number of observations available to fit them, and in which the classical intermittent-demand estimators of Croston [Croston, 1972] and its successors [Syntetos and Boylan, 2005, Teunter et al., 2011] were designed to operate.

Practitioners in this regime receive little guidance. The question they face is concrete and answerable: at what data volume does a deep model stop outperforming a classical statistical baseline, and what governs that threshold? Answering it requires more than re-running existing benchmarks on subsets, because subsetting introduces confounds that are easy to miss. Shorter windows change which series are eligible; injected missingness perturbs the very statistics used to categorize demand patterns; and independent resampling across grid cells discards the statistical power that paired designs provide.

Contributions. This work makes the following contributions.

1. A scarcity sampling protocol (Section 3) that treats history length, series count, missingness rate and missingness shape as controlled experimental factors, stratified by demand pattern, with deterministic manifests for exact reproduction of any grid cell.

2. A dual classification scheme that separates the stable demand-pattern label used for stratification from the label an analyst could actually compute in the scarce regime, and reports the disagreement between them as a quantity of interest rather than as noise.

3. An analysis of confounds specific to scarcity experiments, including a missingness treatment that provably preserves the average inter-demand interval in expectation, and a frame-based eligibility rule that prevents history length from becoming entangled with series longevity.

4. [PENDING EXPERIMENT] An empirical characterization of the crossover point across three datasets, five seeds and the full scarcity grid, together with an analysis of SHAP attribution stability under scarcity.

2 Related Work

Intermittent demand. Croston's method [Croston, 1972] decomposes intermittent series into demand sizes and inter-demand intervals, estimating each separately. Syntetos and Boylan identified and corrected a bias in the original estimator [Syntetos and Boylan, 2005], and Teunter, Syntetos and Babai proposed the TSB variant to handle obsolescence [Teunter et al., 2011]. These methods remain strong baselines precisely because they encode structural assumptions that compensate for scarce data.

Demand categorization. Syntetos et al. [2005] partition series by average inter-demand interval (ADI) and the squared coefficient of variation of demand sizes (CV^2), yielding the smooth, erratic, intermittent and lumpy quadrants used throughout this work. Kostenko and Hyndman [2006] propose an alternative boundary. We adopt the standard cutoffs, since data-driven boundaries would make quadrants incomparable across datasets.

Deep forecasting. N-BEATS [Oreshkin et al., 2020] and the Temporal Fusion Transformer [Lim et al., 2021] represent two dominant families of deep forecaster. Gradient-boosted trees [Ke et al., 2017] occupy a middle ground and were heavily represented among M5 winners [Makridakis et al., 2022]. Our interest is not in any one architecture but in how the ordering among these families changes with data volume.

Evaluation. We follow Hyndman and Koehler [2006] in adopting scaled errors, and Tashman [2000] and Bergmeir and Benítez [2012] in preferring rolling-origin evaluation to a single holdout. Pairwise comparisons use the Diebold–Mariano test [Diebold and Mariano, 1995] with the small-sample correction of Harvey et al. [1997], controlled across the grid by the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995].

The gap. Sample efficiency is studied extensively in general machine learning but rarely in forecasting, where the dominant benchmarks fix the data volume by construction. We are not aware of a controlled study that varies data volume as an experimental factor while holding the series population, the evaluation protocol and the tuning budget constant.

3 The Scarcity Sampling Protocol

3.1 Panel representation

Every dataset is normalized to a long panel of triples (series, t, y) on a regular daily index. The representation makes one distinction that the rest of the protocol depends on:

$$y_{i,t} = \begin{cases} \text{NaN} & \text{the demand process was not observable at } t, \\ 0 & \text{the process was observed and produced zero demand.} \end{cases}$$

Raw retail data conflates these. An item that launches midway through the M5 window carries a long run of leading zeros that are not demand observations at all; the Favorita export omits rows entirely on days without a sale, so absence is ambiguous between a closed store and genuine zero demand. Conflating the two inflates ADI, distorts CV^2, and pushes series into the wrong quadrant. Because the benchmark stratifies on exactly these statistics, the error would not stay local: it would propagate into every reported result. Loaders therefore trim each series to its first strictly positive observation and mark everything prior as unobserved.

3.2 Demand-pattern stratification

For each series we compute ADI as the number of observed periods divided by the number of demand occurrences, and CV^2 as the squared coefficient of variation of the strictly positive demand sizes, and assign a quadrant using the standard cutoffs $\text{ADI} = 1.32$ and $\text{CV}^2 = 0.49$ [Syntetos et al., 2005]. Series with fewer than five demand occurrences are marked insufficient and excluded from the sampling frame, since CV^2 estimated from fewer points is not an estimate of anything.

Classification is performed twice, and this is deliberate. The frame label is computed over full history, is fixed per series, and drives both stratified sampling and analysis grouping. The window label is recomputed inside each sampled window and reflects what an analyst holding only that window could determine. The frame label uses information outside the sampled window, but it is never supplied to any model and therefore leaks nothing into forecasts; it functions purely as a blocking factor, which keeps history length and quadrant membership independent. The disagreement between the two labels is reported per cell as the quadrant drift rate, and is treated as a finding in its own right.

3.3 Missingness and the ADI invariance

Missingness is injected at rates $\{0, 0.05, 0.15, 0.30\}$ under two patterns. MCAR masks observed cells independently. Burst masks contiguous runs with geometric lengths averaging seven days, approximating the failure modes that actually produce gaps in small-business data: a till failing, a store closing, an export job silently breaking for a week. Both are trimmed to the requested rate exactly, so the patterns are compared at identical volumes of absence and differ only in shape. Injection masks only cells that were observed, so the requested rate carries the same meaning regardless of how much native absence a dataset already contains.

The definition of ADI in the preceding subsection is not the conventional one, and the reason is a confound that would otherwise be fatal to the design. Consider counting unobserved periods as part of the inter-demand interval. Under missingness at rate p, the demand count falls in expectation to $(1-p)n$ while the period count remains N, giving

$$\mathbb{E}[\widehat{\mathrm{ADI}}] \;=\; \frac{N}{(1-p)n} \;=\; \frac{\mathrm{ADI}}{1-p},$$

an inflation of 43% at $p = 0.30$. That is far more than enough to drive series across the 1.32 cutoff. The missingness axis of the grid would silently become a second intermittency axis, and the two effects would be inseparable in the results. Restricting the period count to observed periods reduces both numerator and denominator proportionally, so $\mathbb{E}[\widehat{\mathrm{ADI}}] = N/n = \mathrm{ADI}$ and the axes stay orthogonal. Section 5 verifies this empirically.

3.4 Paired seeding

Each cell draws on two independent seeds. The selection seed is derived from the sampling frame alone: dataset, series count, replicate index, frame length, coverage threshold, anchor date and allocation mode. It deliberately excludes history length and every missingness parameter. Consequently the same series appear in every cell along those axes, and a comparison between a 60-day cell and a 730-day cell is a paired comparison on identical series rather than two independent draws. This is a substantial gain in statistical power at no cost, and it removes changes in series composition as a competing explanation for any crossover observed. The injection seed is derived from the complete specification, so each cell still receives independent gaps.

3.5 Frame-based eligibility

A series is eligible if it has at least 80% observed coverage over a fixed 730-day frame, regardless of how much history the cell subsequently hands to the model. Judging eligibility against history_days instead would let short cells draw from a larger pool containing short-lived products while long cells drew only from long-lived ones, entangling history length with series longevity. The cost is some realism, since a genuinely new business does hold short-lived series; the benefit is that an observed crossover is attributable to history length rather than to the population shifting underneath the comparison.

3.6 Allocation and shortfall

Under equal allocation each quadrant receives $n/4$ series, which maximizes power per quadrant and is the purpose of stratifying. Under proportional allocation quadrants receive shares matching the eligible pool, resolved by largest remainder. Proportional allocation exists because it is required: as Section 5 shows, one of our datasets is almost entirely smooth and cannot support equal allocation

Algorithm 1 Drawing one cell of the scarcity grid

Require: dataset D, specification s, frame labels L
frame $\leftarrow$ window of s.frame_days ending at s.end_date
$E \leftarrow \{i : \text{coverage}_i(\text{frame}) \geq s.\text{min\_coverage},\ L_i \neq \text{insufficient}\}$
$r \leftarrow \text{Allocate}(s, \{|E_q|\}_{q \in Q})$ ▷ equal or proportional
$\rho_{\text{sel}} \leftarrow \text{Rng}(\text{Hash}(\text{FrameKey}(s)))$ ▷ excludes history and missingness
$S \leftarrow \bigcup_{q \in Q} \text{DrawWithoutReplacement}(E_q, \min(r_q, |E_q|), \rho_{\text{sel}})$
$W \leftarrow$ last s.history_days of frame, restricted to S
$\rho_{\text{inj}} \leftarrow \text{Rng}(\text{Hash}(s))$ ▷ full specification
$W \leftarrow \text{InjectMissingness}(W, s.\text{rate}, s.\text{pattern}, \rho_{\text{inj}})$
$L^W \leftarrow \text{Classify}(W)$ ▷ window labels, for drift
return W, manifest containing S, r, realized rates, $\text{Drift}(L, L^W)$, panel digest

Table 1: Datasets. Statistics for CTA are measured from the normalized panel. M5 and Favorita figures are as documented by their sources and are not yet independently verified in this pipeline.

Dataset	Domain	Series	Span	Status
M5 [Makridakis et al., 2022]	Retail, hierarchical	30,490	1,941 d	Not yet ingested
Favorita	Grocery, perishables	~200,000	1,684 d	Not yet ingested
CTA	Rail ridership	148	9,312 d	Ingested

at all. When a quadrant holds fewer eligible series than requested, the sampler takes what exists and records the deficit in the manifest rather than topping up from other quadrants, which would silently unbalance the strata the design rests on.

3.7 Reproducibility

Each cell emits a manifest recording the full specification, both derived seeds, window boundaries, requested and realized allocation per quadrant, any shortfall, native and injected and final absence rates, window quadrant counts, the drift rate, the selected series identifiers, and a SHA-256 digest of the resulting panel. Datasets carry two content hashes, one over the raw download and one over the normalized panel, so that an upstream file change and a change in our own normalization code are distinguishable. Results are comparable only when both still match, and overriding a mismatch requires an explicit flag.

4 Experimental Setup

4.1 Datasets

Three datasets span retail and non-retail demand. Table 1 summarizes them; only CTA has been ingested at the time of writing.

CTA rail station entries serve as a non-retail control. A single aggregate ridership series would have been simpler to obtain but useless here, since the protocol samples between 10 and 1,000 series; the station-level panel supports the identical protocol and additionally contains a genuine structural break in 2020.

4.2 Models

Classical baselines comprise seasonal naive, ETS and auto-ARIMA [Hyndman and Khandakar, 2008], with Croston [Croston, 1972] and TSB [Teunter et al., 2011] for intermittent series. The machine-learning tier is LightGBM [Ke et al., 2017] over lag, rolling and calendar features. The deep tier is N-BEATS [Oreshkin et al., 2020] and the Temporal Fusion Transformer [Lim et al., 2021]. All models share a uniform fit/predict interface, an identical feature budget and an identical tuning protocol.

4.3 Metrics and inference

MASE and RMSSE [Hyndman and Koehler, 2006] are primary, WAPE is secondary, and pinball loss is reported for quantile-capable models. MAPE is not used anywhere, as it is undefined on the zero-demand periods that dominate the intermittent regime. Because the protocol injects gaps across the entire window including the evaluation region, unobserved actuals are excluded from every metric rather than coerced to zero; treating them as zeros would systematically favor whichever model degrades most gracefully on missing inputs.

Evaluation is rolling-origin [Tashman, 2000, Bergmeir and Benítez, 2012] with at least five replicates per configuration. Confidence intervals on metric differences are obtained by bootstrap. Pairwise comparisons use Diebold–Mariano [Diebold and Mariano, 1995] with the Harvey–Leybourne–Newbold correction [Harvey et al., 1997], and the false discovery rate across the grid is controlled by Benjamini–Hochberg [Benjamini and Hochberg, 1995].

[PENDING EXPERIMENT] The cross-validation split strategy is not yet fixed. The open tension is that rolling-origin folds on a 60-day window are necessarily few and short, so holding the fold count constant across history lengths and holding the fold size constant are mutually exclusive, and they answer different questions.

5 Protocol Validation

No model has been fit. This section reports only measurements of the sampling protocol itself, which is what is currently implemented and tested.

5.1 The non-retail control has no intermittency

Applying the classifier to the full CTA panel yields the mix in Table 2: essentially every station is smooth. This is not a defect. Rail ridership is a daily continuous process, and the classifier correctly reports that it contains no intermittency to stratify on.

Table 2: Syntetos–Boylan quadrants over full CTA history (148 stations, 2001-01-01 to 2026-06-30).

	Smooth	Erratic	Intermittent	Lumpy	Insufficient
Stations	145	1	1	0	1

The consequence is methodological. Equal allocation degenerates on CTA: a request for 20 series returns 6, because three strata are empty. This is what motivated proportional allocation, and it repositions CTA within the benchmark. It is not a third source of stratified evidence but a contrast case, establishing what the crossover looks like when intermittency is absent. If deep models retain their advantage further down the volume curve on CTA than on M5, that is direct evidence that intermittency rather than sheer data volume is the operative mechanism.

5.2 Quadrant drift is an intermittency effect, not a window-length effect

Table 3 reports drift rates. On real CTA data drift is zero at every history length including 60 days: smooth series are classified robustly from very little data. On synthetic panels containing all four quadrants, drift rises as history shrinks, reaching 7.5% at 60 days.

Taken together these sharpen the claim available to us. Short histories do not mislabel series indiscriminately; they mislabel specifically those series whose CV^2 rests on a handful of demand occurrences, which are exactly the series for which the choice of forecasting method matters most.

Table 3: Quadrant drift rate, the fraction of sampled series whose quadrant changes when estimated from the window rather than from full history. Synthetic panels use 25 series per quadrant with cleanly separated generators and therefore represent a lower bound; real catalogues sit closer to the cutoffs.

History (days)	60	120	365	730
CTA, no missingness	0.000	0.000	0.000	0.000
CTA, 30% burst	0.000	0.000	0.000	0.000
Synthetic, no missingness	0.075	0.025	0.000	0.000
Synthetic, 30% MCAR	0.075	0.000	0.000	0.000

5.3 The ADI invariance holds empirically

Section 3.3 argued that restricting the period count to observed periods leaves ADI unchanged in expectation under missingness. The final two rows of Table 3 provide the check: at 60 days of history, drift is 0.075 both without missingness and at 30% MCAR. Injecting gaps into nearly a third of the panel does not reshuffle quadrant membership. Under the conventional definition the same manipulation would have inflated ADI by 43%.

5.4 A defect visible only in real data

The CTA feed contains 1,236 rows that duplicate a station–day, all within a 41-day window in mid-2011, each reporting a different ride count for the same day. The affected share is 0.09% of the panel, with median disagreement 0.3% and maximum 20%. No principled basis exists for preferring either figure, and averaging them would fabricate an observation that was never recorded, so the affected station–days are marked unobserved. We note this less for its magnitude than for its provenance: it was invisible to a test suite built on synthetic panels and surfaced within seconds of contact with the real feed.

6 Results

[PENDING EXPERIMENT] This section will report the crossover analysis across the full scarcity grid: metric surfaces by history length and series count for each model family, stratified by demand-pattern quadrant; bootstrap confidence intervals on metric differences; Diebold–Mariano tests with multiplicity control; and SHAP attribution stability across seeds and scarcity regimes. None of this has been run. Deliberately left empty rather than estimated, projected or illustrated.

7 Limitations

Several limitations are known in advance and are properties of the design rather than oversights.

Frame-based eligibility trades realism for attribution. Requiring 730 days of coverage excludes precisely the short-lived products a genuinely new business would carry. The benchmark therefore measures the effect of showing a model less history, not the effect of a business being young. These are different questions and only the first is cleanly identifiable.

Stratification uses out-of-window information. The frame label could not be computed by a practitioner holding only the sampled window. It is never given to a model, so no forecast is contaminated, but the strata are in this sense idealized. The drift rate is reported precisely to quantify what that idealization conceals.

The non-retail control is not stratified. CTA is almost entirely smooth and contributes evidence only about the smooth quadrant.

Injected missingness is not real missingness. MCAR and geometric bursts are tractable idealizations. Real gaps correlate with demand itself: stockouts suppress recorded sales precisely when demand is high, which is missingness not at random [Little and Rubin, 2019]. Our design cannot capture that dependence and will therefore, if anything, understate the difficulty of the real problem.

8 Conclusion

[PENDING EXPERIMENT] Conclusions require results. This section is intentionally empty.

Reproducibility

All code, configurations and design rationale are available at https://github.com/aryanshirodkar15/scarce-forecasting. Raw data is not redistributed; download scripts and content hashes are provided so that any reported figure can be traced to the exact bytes that produced it. Every grid cell is reconstructible from its manifest.

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